

Lewis Lehe

ASSOCIATE PROFESSOR · CIVIL AND ENVIRONMENTAL ENGINEERING
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Education

UC Berkeley

PHD CIVIL AND ENVIRONMENTAL ENGINEERING

- Advisor: Dr. Carlos Daganzo
- Concentration: Transportation Engineering

Berkeley, CA

2013-2016

UC Berkeley

MS CIVIL AND ENVIRONMENTAL ENGINEERING

- Concentration: Transportation Engineering

Berkeley, CA

2012 - 2013

University of Leeds

MA TRANSPORT ECONOMICS

- Thesis: Income Effects in Discrete Choice Models

Leeds, United Kingdom

2011 - 2012

University of Pittsburgh

B.PHIL. MATHEMATICS-ECONOMICS

Pittsburgh, PA

2005 - 2010

Publications

(advised students' names in bold)

Shen, S., Zhang, Y., **Qiam, S.**, Crandall, P., Krein, P., Lehe, L.J., Kontou, E. and Ouyang, Y. (2026) "How Ubiquitous Slow Charging Saves Transportation Electrification Costs in Metropolitan Areas." *2026 IEEE Transportation Electrification Conference and Expo and Electric Aircraft Technologies Symposium (ITEC+EATS)*

Ahammed, D.T., Shih, B.J. and Lehe, L.J. (2026) "A GTFS-Based Dataset for the US Intercity Bus Network." *Findings*

Qiam, S. and Lehe, L.J. (2025) "Off-street Parking in 15 US Cities." *Findings*

Lehe, L.J. **Pandey, A.** (2025) "Equilibrium horizontal queues and a tolling paradox" *Transportation Research Part B: Methodological*

Lehe, L.J., **Devunuri, S.**, Rondan, J., **Pandey, A.**, Vignon, D. (2025) "Taxation of ridehailing in the United States" *Transport Policy*

Qiam, S. and **Devunuri, S.** and Lehe, L.J. (2025) "A Pipeline and NIR-Enhanced Dataset for Parking Lot Segmentation." *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*

Pandey, A. and Lehe, L.J. (2025) "Determinants of US Transit Bus Prices." *Findings*

Lehe, L.J. **Pandey, A.** (2025) "Market size and fare free public transport in theory" *Transportation Research Part B: Methodological*

Devunuri, S. and Lehe, L.J. (2025) "TransitGPT: a generative AI-based framework for interacting with GTFS data using large language models." *Public Transport*

Pandey, A. and Lehe, L.J. (2024) "Bus stop spacing with heterogeneous trip lengths and elastic demand." *Transportation Research Part B: Methodological*

Devunuri, S. and Lehe, L.J. (2024) "A survey of errors in GTFS Static feeds in the United States." *Findings*

Pandey, A. and Lehe, L.J. (2024) "Local Stability of Traffic Equilibria in an Isotropic Network." *Transportation Research Part B: Methodological*

Lehe, L.J. and **Pandey, A.** (2024) "A bathtub model of transit congestion." *Transportation Research Part B: Methodological*

- Devunuri, S., Qiam, S.,** and Lehe, L.J. (2024) “ChatGPT for GTFS: benchmarking LLMs on GTFS semantics and retrieval.” *Public Transport*
- Devunuri, S.,** Lehe, L.J., **Qiam, S., Pandey, A.** and Monzer, D. (2024) “Bus stop spacing statistics: theory and evidence.” *Journal of Public Transportation*
- Devunuri, S.** and Lehe, L.J. (2024) “GTFS Segments: A Fast and Efficient Library to Generate Bus Stop Spacings.” *Journal of Open Source Software*
- Pandey, A.** and Lehe, L.J. (2024) “Congestive mode-switching and economies of scale on a bus route.” *Transportation Research Part B: Methodological*
- Liu, H., **Devunuri, S.,** Lehe, L.J., Gayah, VV., (2023) “Scale Effects in Ridesplitting: A Case Study of the City of Chicago.” *Transportation Research Part A: Policy and Practice*
- Lehe, L., **Pandey, A.,** (2022) “Taxi service with heterogeneous drivers and a competitive medallion market.” *Journal of Urban Economics*
- Lehe, L.J., **Devunuri, S.,** (2022) “Large Elasticity at Introduction.” *Research in Transportation Economics*
- Lehe, L.J., **Devunuri, S.,** Rondan, J., **Pandey, A.,** (2021) *Taxation of Ridehailing.* Illinois Center for Transportation.
- Lehe, L.J., Gayah, VV., **Pandey, A.,** (2021) “Increasing Returns to Scale in Carpool Matching: Evidence from Scoop.” *Findings*
- Lehe, L.J. (2020) “Winners and losers from road pricing with heterogeneous travelers and a mixed-traffic bus alternative.” *Transportation Research Part B: Methodological*
- Hu, S., Zhou, Q., Li, J., Wang, Y., Roncoli, C., Zhang, L., Lehe, L.J., (2022) “High Time-Resolution Queue Profile Estimation at Signalized Intersections Based on Extended Kalman Filtering.” *IEEE Transactions on Intelligent Transportation Systems*
- Pandey, A.,** Lehe, L.J., Monzer, D., (2021) “Distributions of Bus Stop Spacings in the United States.” *Findings*
- Lehe, L.J., **Pandey, A.,** (2020) “Hyperdemand: A static traffic model with backward-bending demand curves.” *Economics of Transportation*
- Lehe, L.J. (2019). “Downtown congestion pricing in practice.” *Transportation Research Part C: Emerging Technologies*
- Lehe, L.J. (2018) “Minimum parking requirements and housing affordability.” *Journal of Transport and Land Use*
- Lehe, L.J. (2017) “Feedback and the use of land for parking.” *Journal of Transport and Land Use*
- Lehe, L.J., (2017) “Downtown tolls and the distribution of trip lengths.” *Economics of Transportation*
- Daganzo, C.F., Lehe, L.J., Argote-Cabanero, J., (2018) “Adaptive offsets for signalized streets.” *Transportation Research Part B: methodological*
- Daganzo, C.F., Lehe, L.J., (2016) “Traffic flow on signalized streets.” *Transportation Research Part B: Methodological*

Awards, Fellowships, & Grants _____

2025	CEE Teaching Excellence Award , University of Illinois at Urbana-Champaign Civil and Environmental Engineering Department	
Fall 2024	Teachers Ranked as Excellent by their Students , CEE 310 Transportation Engineering	
Spring 2024	Teachers Ranked as Excellent by their Students , CEE 419 Transportation Economics	
Fall 2023	Teachers Ranked as Excellent by their Students , CEE 310 Transportation Engineering	
Spring 2022	Teachers Ranked as Excellent by their Students , CEE 498TE Transportation Economics	
Fall 2021	Teachers Ranked as Excellent by their Students , CEE 310 Transportation Engineering	
Spring 2020	Teachers Ranked as Excellent by their Students , CEE 498TE Transportation Economics	
2024-2029	NSF CAREER: The Choice of Free vs. Priced for Transportation Systems , National Science Foundation - Civil, Mechanical and Manufacturing Innovation	\$519,000
2021-2024	Economics and Control for Next-Generation Carpooling (w/ Dr. Vikash Gayah at Penn State) , National Science Foundation - Civil, Mechanical and Manufacturing Innovation	\$400,000
2021-2022	Mobility On-demand Lab (w/ Peter Christensen and Adam Osman) , Discovery Partners Institute	\$ 120,000
2021-2022	The Last Mile Problem - a Grand Transportation Challenge (with Peter Christensen and Adam Osman) , National Bureau of Economic Research	\$ 75,000
2020-2021	Taxation of Ridesharing , Illinois Center for Transportation	\$ 160,000

Teaching Experience _____

2026	CEE 416: Traffic Capacity Analysis ,
2025	CEE 418: Public Transportation ,
2020+	CEE 419: Transportation Economics ,
2019+	CEE 310: Transportation Engineering ,

Outreach & Professional Development _____

SERVICE

2024+	International Transportation Economics Association , Communications Officer
2024+	International Transportation Economics Association , Scientific Committee
2022+	Institute for Transportation Engineers Student Chapter , Faculty Advisor
2019+	Findings Press , Treasurer and Editor
2019-2025	UIUC Faculty Senate , Senator